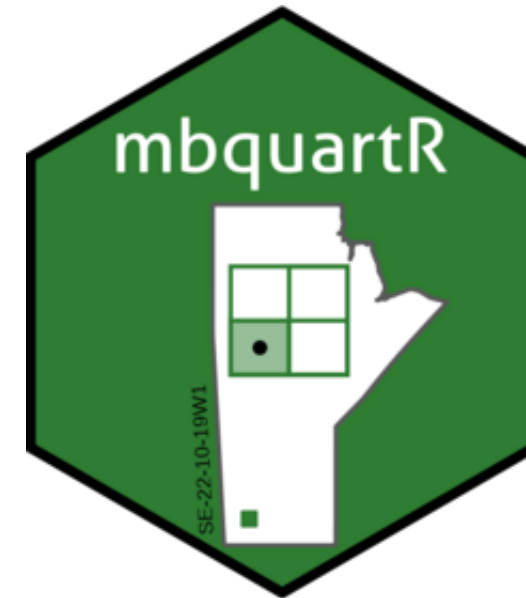


mbquartR: An R Package for Working with Manitoba Quarter Sections

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Brandon University



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 [alex-koiter](https://github.com/alex-koiter)
 [Alex-Koiter](https://www.linkedin.com/in/Alex-Koiter)



[Visit the mbquartR website](https://mbquartR.com)

What is a quarter section?

- Land unit measuring 160 acres (~64.8 ha), representing a quarter of a square mile
- Originates from the Dominion Land Survey system
 - European settlement and colonization of Western Canada
- Explains why most farms are square!



What is a quarter section?

- A legal land description for a quarter section consists of four values separated by a “-”

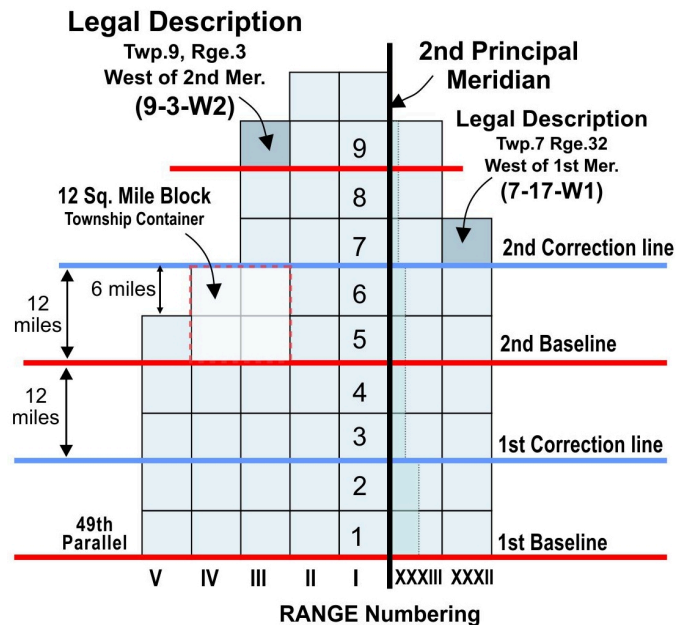
1. Quarter Section (SW)

3. Township (8)

2. Section (9)

4. Range (6E1)

- SW-9-8-6E1:** Southwest Quarter of Section 9, Township 8, Range 6 East of the 1st Meridian

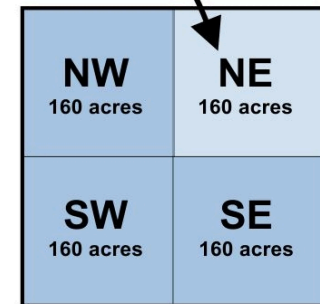


TOWNSHIP 9, RANGE 3 W2 or (9-3-W2)



SECTION 25
640 acres - 1 square mile

1/4 SECTION Legal Description
160 acres **NE25-9-3-W2**



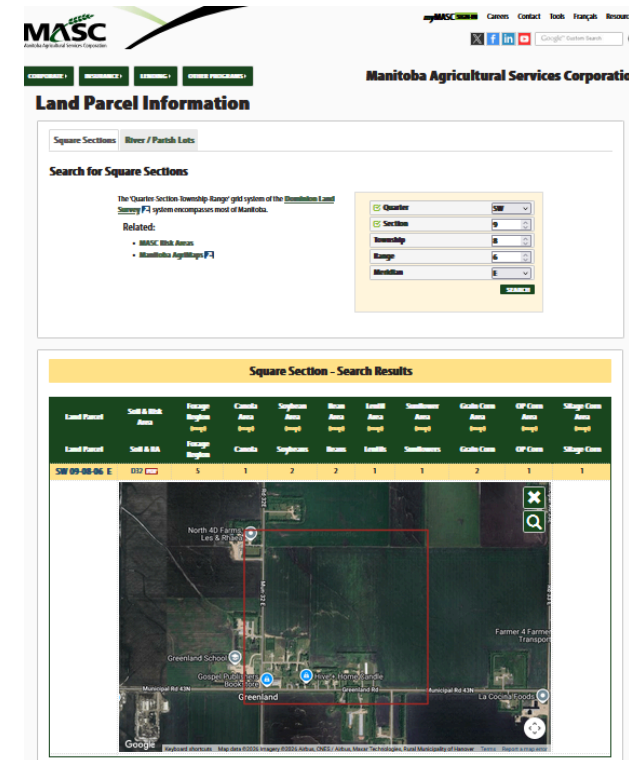
Why are they used?

- Commonly used as a georeference before the widespread use of GPS
 - Archived samples and data
- Still commonly used in rural areas
 - A bit like a street address



Why are they difficult to use?

- Lack of familiarity with the system
- Google Maps can't find SW-9-8-6E1 so I don't know how to get there!
- Online resources do exist to locate to help find them
 - Fee based
 - No batch processing
 - Do not provide coordinates
 - Can not search by coordinates



How it all started

- An email

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Dr. Steve Crittenden:

Hi Alex,

I need to locate coordinates for a bunch of quarter sections.

There must be an R package for this, right?

How it all started

- An email

Dr. Steve Crittenden:

Hi Alex,

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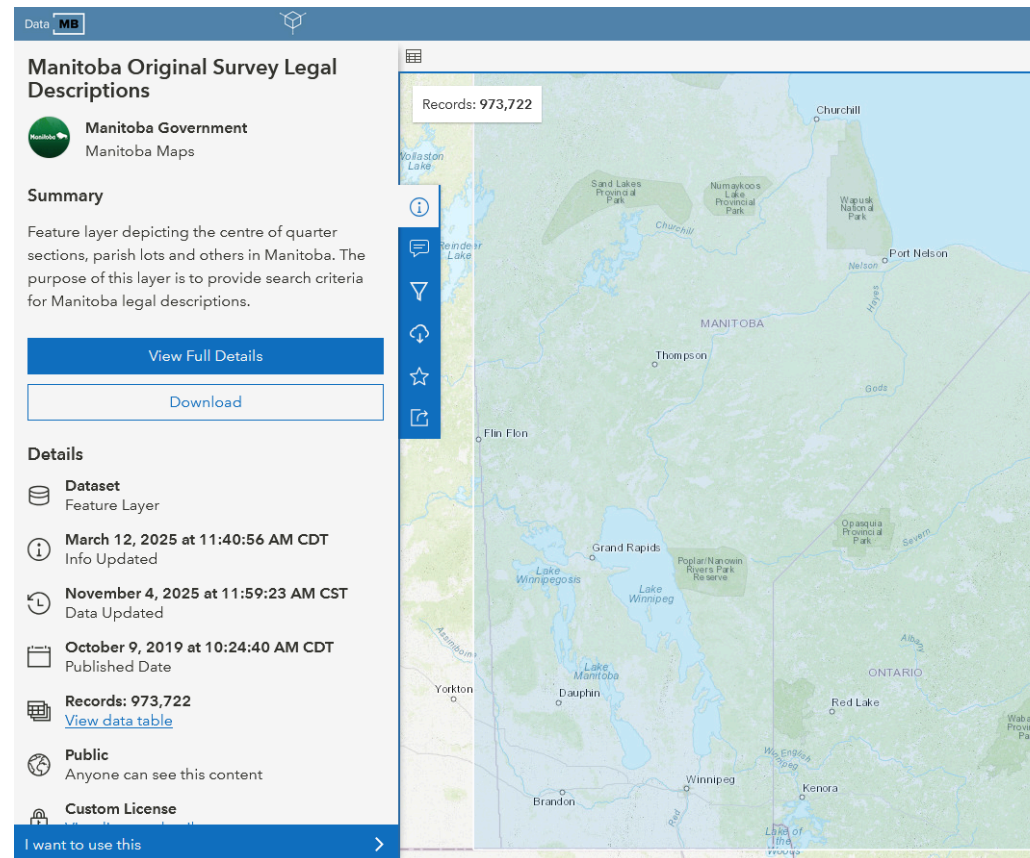
There must be an R package for this, right?

- But there wasn't! I was surprised and wanted to help

How it all started

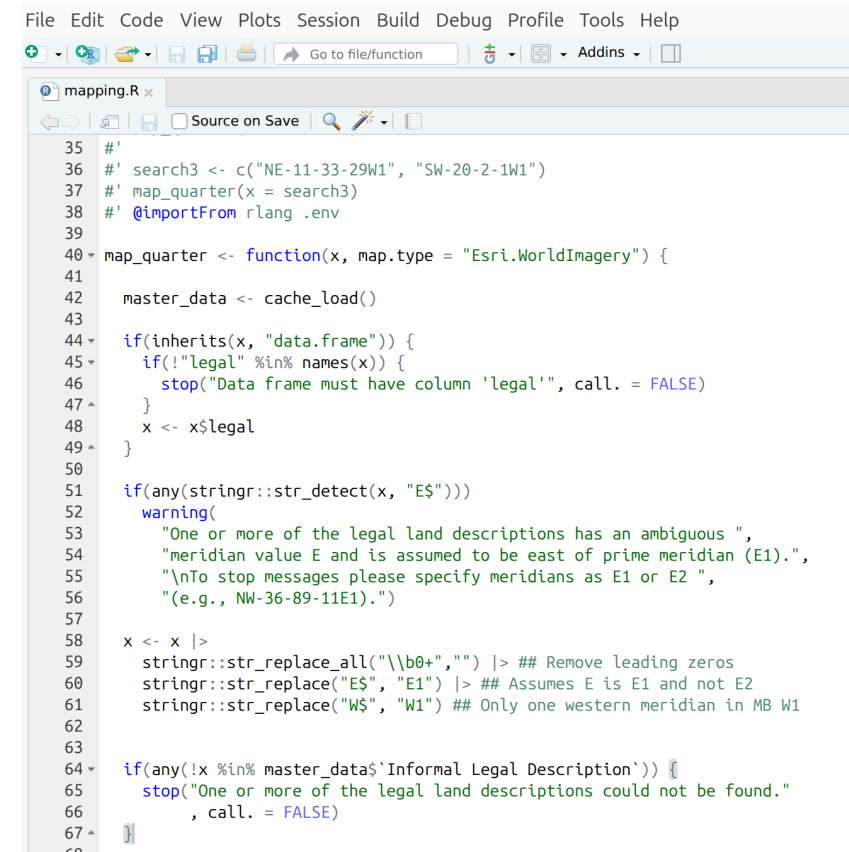
- Data was publicly available through [Data MB](#)

1. Wrote a script that downloaded the database and searched the database



How it all started

- Data was publicly available through [Data MB](#)
1. Wrote a script that downloaded the database and searched the database
 2. Which turned into a series of functions



```
35 #'
36 #' search3 <- c("NE-11-33-29W1", "SW-20-2-1W1")
37 #' map_quarter(x = search3)
38 #' @importFrom rlang .env
39
40 map_quarter <- function(x, map.type = "Esri.WorldImagery") {
41
42   master_data <- cache_load()
43
44   if(inherits(x, "data.frame")) {
45     if(!"legal" %in% names(x)) {
46       stop("Data frame must have column 'legal'", call. = FALSE)
47     }
48     x <- x$legal
49   }
50
51   if(any(stringr::str_detect(x, "E$")))
52     warning(
53       "One or more of the legal land descriptions has an ambiguous ",
54       "meridian value E and is assumed to be east of prime meridian (E1).",
55       "\nTo stop messages please specify meridians as E1 or E2 ",
56       "(e.g., NW-36-89-11E1).")
57
58   x <- x |>
59     stringr::str_replace_all("\\b0+", "") |> ## Remove leading zeros
60     stringr::str_replace("E$", "E1") |> ## Assumes E is E1 and not E2
61     stringr::str_replace("W$", "W1") ## Only one western meridian in MB W1
62
63
64   if(any(!x %in% master_data$`Informal Legal Description`)) {
65     stop("One or more of the legal land descriptions could not be found.",
66         , call. = FALSE)
67   }
68 }
```

How it all started

- Data was publicly available through [Data MB](#)
1. Wrote a script that downloaded the database and searched the database
 2. Which turned into a series of functions
 3. Ultimately into an R package

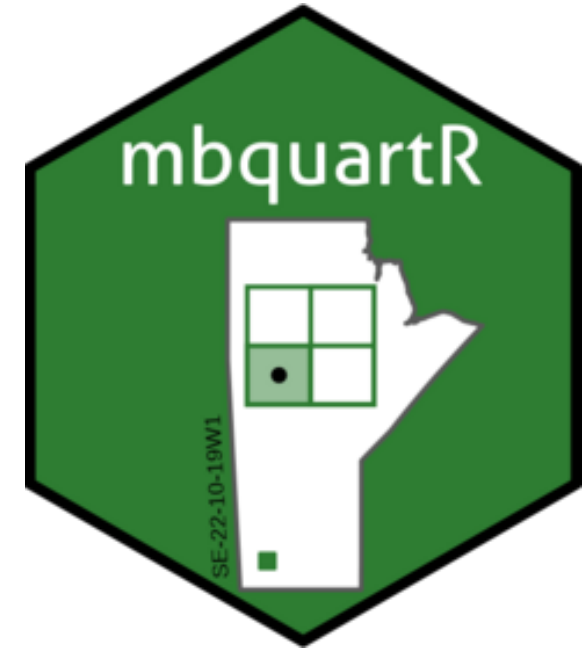
```
File Edit Code View Plots Session Build Debug Profile Tools Help
+ - - - - - Go to file/function + - - - - - Addins - [ ]

mapping.R x
[ ] Source on Save [ ] [ ] [ ] [ ] [ ]
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Why use mbquartR?

Free

- Free and open-source software (FOSS)



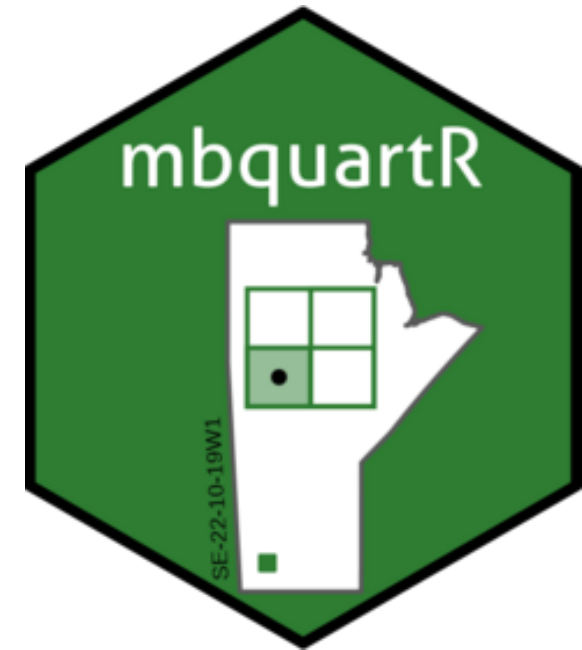
Why use mbquartR?

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Fast and Easy

- One line of code to download data
- One line of code to search
- One line of code to map



Why use mbquartR?

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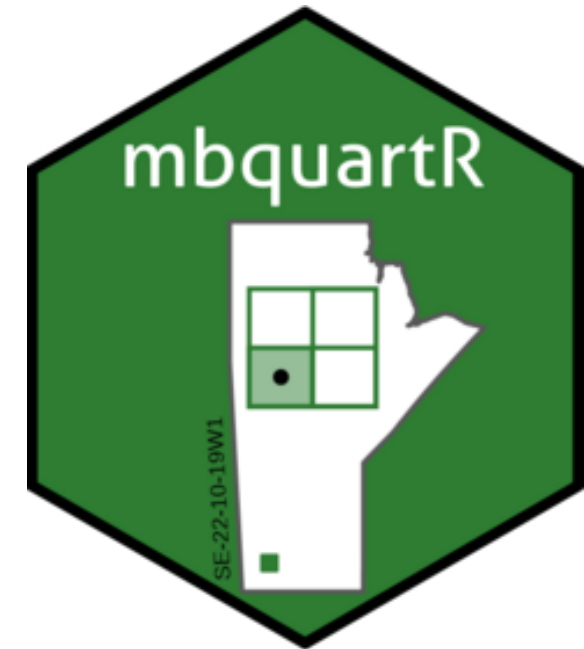
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Fast and Easy

- One line of code to download data
- One line of code to search
- One line of code to map

Integrate into R work flows

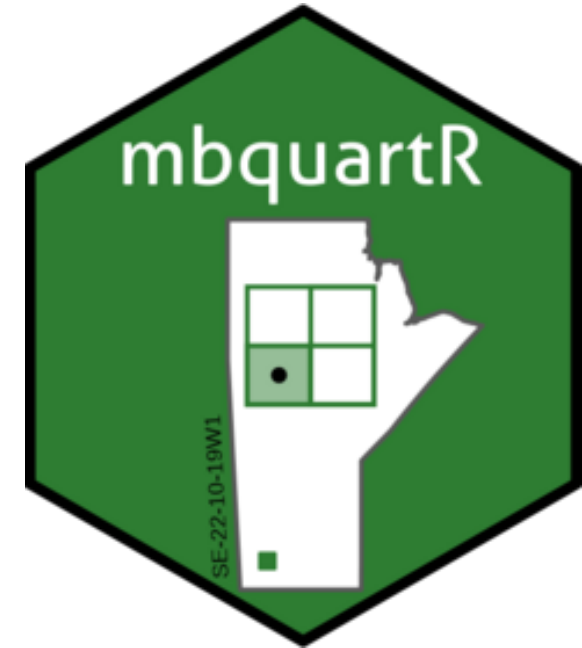
- Coordinates can be used to link multiple datasets
- Working with older archived soil samples or data



Why use mbquartR?

Reproducible!

- Scripts provide a record of actions
- Just note the `mbquartR` version
 - `packageVersion(mbquartR)`
- Hard to document mouse clicks or website searches



mbquartR

Installing mbquartR

```
1 install.packages("mbquartR", repos = "https://ropensci.r-universe.dev")
```

- Available through the R-universe



mbquartR

Downloading the data

```
1 quarters_dl()
```

- Downloads the data from Data MB
 - If website is not available it downloads an archived version
 - Data is not likely to change over time
 - More than 900,000 parcels of land!



mbquartR

Search by legal name

```
1 search_legal(x = c("NE-11-33-29W", "SW-20-2-1E"))
```

mbquartR

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Output

Warning in search_legal(x = c("NE-11-33-29W", "SW-20-2-1E")): One or more of the legal land descriptions has an ambiguous meridian value E and is assumed to be east of prime meridian (E1).

To stop messages please specify meridians as E1 or E2 (e.g., NW-36-89-11E1).

A tibble: 2 × 4

	legal	type	long	lat
	<chr>	<chr>	<dbl>	<dbl>
1	NE-11-33-29W1	Quarter	-101.	51.8
2	SW-20-2-1E1	Quarter	-97.4	49.1

mbquartR

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1	NE-11-33-29W1	Quarter	-101.	51.8
2	SW-20-2-1E1	Quarter	-97.4	49.1

- Spelling and order matters
- You can include leading zeroes (e.g., NE-01-012-12E1) or not (e.g., NE-1-12-12E1)
- Unless you specify it assumes you are using the E1 meridian not E2

mbquartR

Search by coordinates

```
1 search_coord(long = c(-101.4656, -99.99768), lat = c(51.81913, 49.928926))
```

mbquartR

Search by coordinates

```
1 search_coord(long = c(-101.4656, -99.99768), lat = c(51.81913, 49.928926))
```

Output

```
# A tibble: 2 × 7
  legal      type    long   lat long_user lat_user dist
  <chr>      <chr>  <dbl> <dbl>   <dbl>   <dbl>  [m]
1 NE-11-33-29W1 Quarter -101.   51.8   -101.    51.8  29.9
2 NW-15-11-19W1 Quarter -100.   49.9   -100.    49.9 211.
```

mbquartR

Mapping

```
1 map_quarter(x = c("NE-11-33-29W1", "SW-20-2-1W1"), map.type = "Esri.WorldImagery")
```

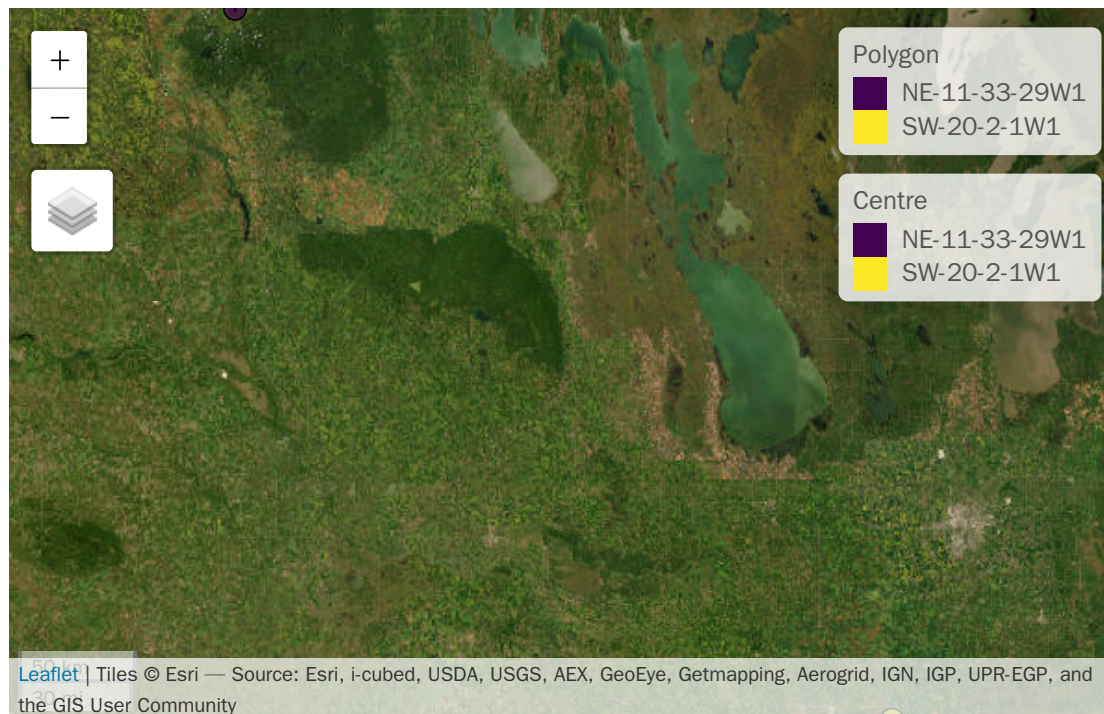
mbquartR

Mapping

```
1 map_quarter(x = c("NE-11-33-29W1", "SW-20-2-1W1"), map.type = "Esri.WorldImagery")
```

Output

Polygons for quarter sections are approximate and are only meant as a visual aid.



mbquartR

Documentation

- All functions have help documentation

```
1 ?map_quarter()
```

map_quarter {mbquartR}

R Documentation

Quickly find MB quarter sections on a map

Description

Map the output of `search_legal()` or `search_coord()`, similar data frame, or a vector of legal land descriptions onto an interactive Leaflet map.

Usage

```
map_quarter(x, map.type = "Esri.WorldImagery")
```

Arguments

`x` Output from `search_legal()` or `search_coord()`, similar data frame, or a vector of legal land descriptions.

`map.type` Base map: see <https://leaflet-extras.github.io/leaflet-providers/preview/> for available options

Details

Map displays the centre point and estimates the boundaries for quarter sections.

Map locations using a data frame which must include a column named `legal` containing legal land descriptions.

Map locations using a vector of legal land descriptions.

Value

mbquartR

Documentation

- Website
 - Vignettes
 - Examples

 mbquartR 0.1.0

Search for

Get started Reference Articles ▾ Changelog

mbquartR



  Peer Reviewed

The goal of `mbquartR` is to provide an easy way to download the Manitoba Original Survey Legal Descriptions data from [DataMB](#) and then to locate quarter sections, and other land division types, in the province of Manitoba. You can search by legal land description (e.g., NE-11-33-29W1) or by lat/long coordinates (e.g., -101.4656, 51.81913). There is also a convenient map function that plots the centres and outlines (polygons) of the quarter sections on a map.

The Manitoba Original Survey Legal Descriptions data set covers the entire province of Manitoba and there are over 900,000 parcels of land that have a legal land description. `mbquartR` was created for those who work with geospatial data in Manitoba, particularly those who are working with rural or farm parcels of land where the legal land description is commonly used as the method of identifying the location. `mbquartR` also allows users to quickly go back and forth between geographic coordinates and the legal land description. Most mapping applications (e.g., Google Maps) can not find or route to a legal land description, but you can with coordinates!

Checkout the [mbquartR website](#) for vignettes/code examples.

What is a quarter section?

A quarter section in Manitoba is a land unit measuring 160 acres (~64.8 ha), representing a quarter of a square mile. It originates from the [Dominion Land Survey system](#), introduced in the late 19th century to organize the European settlement and colonization of Western Canada. The grid system covers most of the province of Manitoba and organizes land into a hierarchy of quarter sections, sections, townships, and ranges.

Links

[Browse source code](#)
[Report a bug](#)

License

[Full license](#)
[MIT](#) + file [LICENSE](#)

Community

[Contributing guide](#)

Citation

[Citing mbquartR](#)

Developers

Alex Koiter
Author, maintainer 
[More about authors...](#)

R-universe

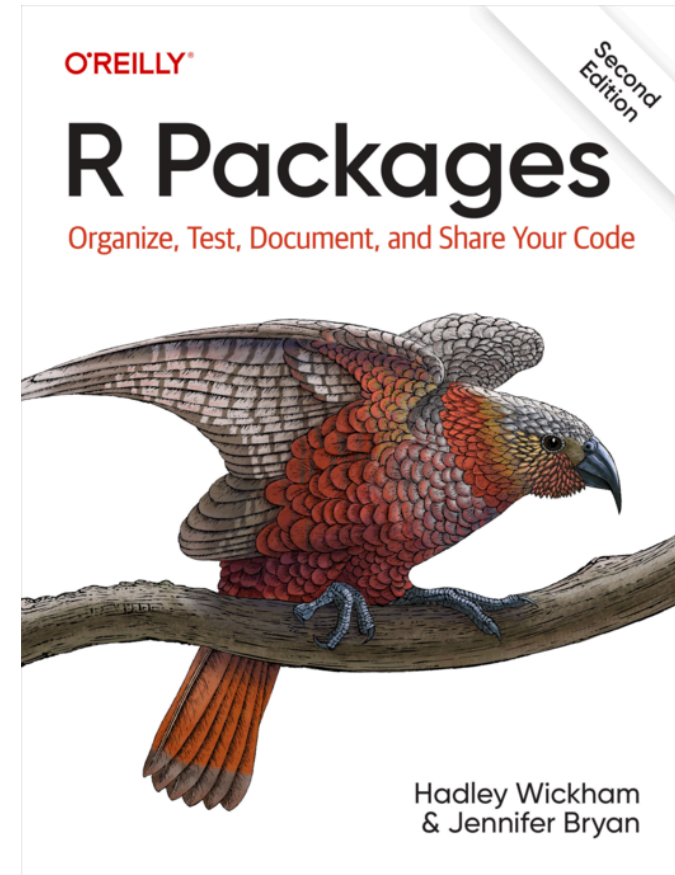
 0.1.0

Software Peer-Review

 Peer Reviewed

Package development

- Lots of great online resources
 - [R Packages \(2e\)](#)
 - [Happy Git and GitHub for the useR](#)
 - [rOpenSci Dev Guide](#)
- Fantastic developer tools/packages
 - `usethis`
 - `testthat`
 - `covr`



Hadley Wickham and Jennifer Bryan

Package development

- Asked for lots of help along the way (thanks Steffi!)
- Looked at other examples (open source is awesome!)
- Tried really hard to break the package and anticipate unusual usage
- Helpful messages, warnings, and errors
- Useful package documentation



Allison Horst

I learned a lot throughout this process!

rOpenSci Software Review

- The goal of the review is to improve the quality, reliability, and maintainability of scientific R packages
- Entire process is open and transparent ([my review](#))
- [Code of conduct](#) ensured a nice experience
- Pre-Submission Enquiry
 - Avoids putting in the work only to get a desk rejection
- Easy to follow [guidelines](#)



rOpenSci Software Review

- My reviewers had excellent suggestions
- I was able to have a back and forth conversation with the reviewers
- Blog post about my experience ([English](#) / [Español](#))



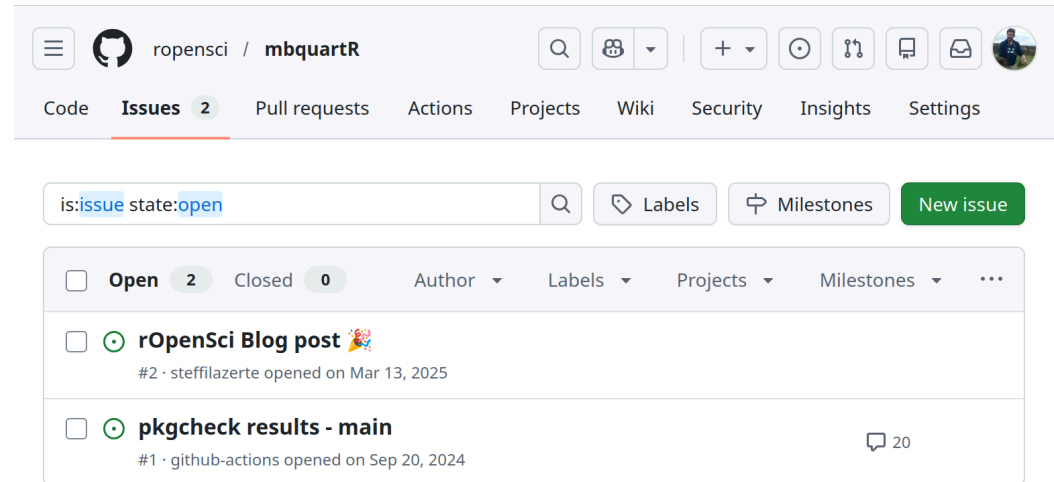
rOpenSci Software Review

- My reviewers had excellent suggestions
- I was able to have a back and forth conversation with the reviewers
- Blog post about my experience ([English](#) / [Español](#))
- Once accepted, online code was transferred to rOpenSci
- I retain authorship and ownership
 - rOpenSci continues to provide support



Want to contribute?

- Open an issue on the [mbquartR repository](#)
 - Report a bug
 - Request a feature
 - Ask a question



Want to contribute?

- Open an issue on the [mbquartR repository](#)
 - Report a bug
 - Request a feature
 - Ask a question
- Read the [contributing guide](#)
 - Bug fixes
 - Feature implementation

 mbquartR 0.1.0

Contributing to mbquartR

Source: [.github/CONTRIBUTING.md](#)



All suggestions are welcomed, big and small, on how to make this package more robust, functional, and user friendly. Please read the contributing guide below and follow it the best you can.

By participating in this project, you agree to abide by the following [code of conduct](#).

Fixing typos

You can fix typos, spelling mistakes, or grammatical errors in the documentation directly using the GitHub web interface, as long as the changes are made in the *source* file. This generally means you'll need to edit [roxygen2 comments](#) in an `.R`, not a `.Rd` file. You can find the `.R` file that generates the `.Rd` by reading the comment in the first line.

Bigger changes

If you want to make a bigger change, it's a good idea to first file an issue and make sure the author agrees that it's needed. If you've found a bug, please file an issue that illustrates the bug with a minimal [reprex](#) if you can (this will also help you write a unit test, if needed).

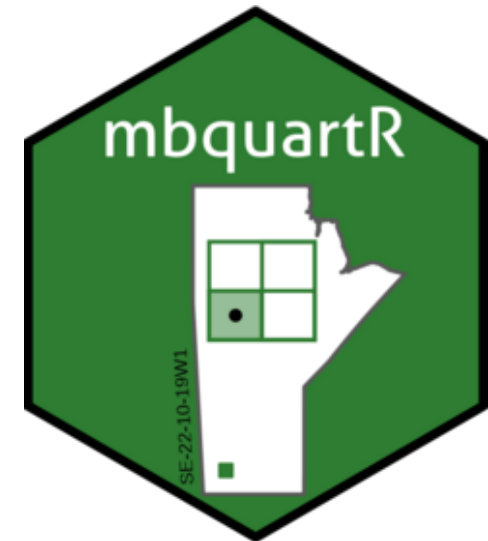
Pull request process

- Fork the package and clone onto your computer. If you haven't done this before, we recommend using `usethis::create_from_github("alex-koiter/mbquartR", fork = TRUE)`.
- Install all development dependencies with `devtools::install_dev_deps()`, and then make sure the package passes R CMD check by running `devtools::check()`. If R CMD check doesn't pass cleanly, it's a good idea to ask for help before continuing.

mbquartR: Finding MB quarter sections made easier!

Want to learn more?

 alexkoiter.ca
 koitera@brandonu.ca
 [alex-koiter](https://github.com/alex-koiter)
 [alex-koiter](https://bitbucket.org/alex-koiter)
 @alex_koiter@mstdn.ca
 [Alex-Koiter](https://www.linkedin.com/in/Alex-Koiter)
 [@alex_koiter](https://twitter.com/alex_koiter)



Slides created with [Quarto](#)

Updated 2026-02-03